

Cheat Sheet: Analysis of 16s Data

This guide summarizes the key steps, tools, and concepts from the Galaxy Project's "Analyses of 16s data - The global picture" tutorial.

Key Concepts

- **Amplicon Sequencing (e.g., 16S rRNA):** A targeted approach that sequences a specific marker gene (like 16S rRNA for bacteria) to determine **which microbes are present** (taxonomic profile). It's good for "who is there?"
 - **Shotgun Sequencing:** A non-targeted approach that sequences all genetic material (genomes) in a sample to determine both **which microbes are present** (taxonomy) and **what they are capable of doing** (functional profile). It answers "who is there?" and "what can they do?"
 - **OTU (Operational Taxonomic Unit):** A cluster of similar sequences, often grouped at 97% similarity, used as a proxy for a bacterial species or genus.
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Amplicon Data Analysis (16S rRNA) Workflow

This workflow uses the **mothur** tool suite to analyze 16S rRNA data and identify the taxonomic composition of samples.

1. Data Import & Preparation

1. **Import Data:**
 - **Tool:** Upload Manager
 - **Purpose:** Import sequence data. The tutorial uses FASTA files from Zenodo links (e.g., SRR531818_pampa.fasta, SRR651839_anguil.fasta).
 - **Action:** Rename datasets for clarity (e.g., "pampa" and "anguil").
2. **Merge Samples:**
 - **Tool:** Merge.files
 - **Purpose:** Combine all individual sample FASTA files into one single FASTA file for multisample analysis.
3. **Create Group File:**
 - **Tool:** Make.group
 - **Purpose:** Create a file that maps each sequence (read name) to its original sample (group name).
 - **Key Parameters:**

- Method...: "Manually specify..."
- fasta: Select an original sample file (e.g., SRR531818_pampa.fasta).
- group: Assign a name (e.g., "pampa").
- Repeat for all samples.

2. Optimization & Initial QC

1. Dereplicate Sequences:

- **Tool:** Unique.seqs
- **Purpose:** Reduces computational load by finding identical sequences (dereplication) and collapsing them. Creates a representative sequence for each unique type.
- **Outputs:** A .fasta file (unique sequences) and a .names file (lists which original reads are represented by each unique sequence).

2. Count Sequences:

- **Tool:** Count.seqs
- **Purpose:** Creates a count table to track how many times each unique sequence appears in each sample (group).
- **Key Parameters:**
 - name: The .names file from Unique.seqs.
 - group: The .group file from Make.group.

3. Quality Control & Filtering

1. Summarize Data:

- **Tool:** Summary.seqs
- **Purpose:** Get summary statistics about the sequences (e.g., length, ambiguities) to decide on filtering parameters.
- **Key Parameters:** fasta (from Unique.seqs), count (from Count.seqs).

2. Screen Sequences:

- **Tool:** Screen.seqs
- **Purpose:** Filter and remove low-quality sequences based on length, ambiguous bases, and homopolymers.
- **Key Parameters:**
 - fasta: The .fasta file from Unique.seqs.
 - count: The count table from Count.seqs.
 - minlength: e.g., 225
 - maxlength: e.g., 275
 - maxambig: e.g., 0
 - maxhomop: e.g., 8 (removes sequences with homopolymers of 9 or more).

4. Sequence Alignment

1. Import Reference:

- **Tool:** Upload Manager
- **Purpose:** Import the reference alignment database (e.g., silva.v4.fasta).

2. Align Sequences:

- **Tool:** Align.seqs
- **Purpose:** Align your "good" sequences to the reference alignment (e.g., SILVA).
- **Key Parameters:**
 - **fasta:** The good.fasta file from Screen.seqs.
 - **reference:** The silva.v4.fasta file.
 - **flip:** Yes (to check reverse-complement).

3. Screen Aligned Sequences:

- **Tool:** Screen.seqs
- **Purpose:** Remove sequences that did not align well to the target region.
- **Key Parameters:**
 - **fasta:** The aligned .fasta file from Align.seqs.
 - **start:** e.g., 3080 (based on Summary.seqs output).
 - **end:** e.g., 13424 (based on Summary.seqs output).
 - **count:** The count table from the *previous* Screen.seqs step.

4. Filter Alignment:

- **Tool:** Filter.seqs
- **Purpose:** Remove alignment columns that are only gaps ('.') to make sequences overlap the same region.
- **Key Parameters:**
 - **fasta:** The good.fasta from the *aligned* Screen.seqs step.
 - **trump:** . (removes columns containing only '.' characters).

5. Taxonomic Extraction & OTU Clustering

1. Pre-cluster Sequences:

- **Tool:** Pre.cluster
- **Purpose:** Further denoises the data by merging sequences that are very similar (likely due to sequencing error).
- **Key Parameters:**
 - **fasta:** The filtered .fasta file from Filter.seqs.
 - **name file or count table:** The count table from the last Screen.seqs step.
 - **diffs:** 2 (allows up to 2 nucleotide differences).

2. Import Taxonomy:

- **Tool:** Upload Manager
- **Purpose:** Import the reference taxonomy files (e.g., trainset16_022016.pds.fasta)

and trainset16_022016.pds.tax).

3. Classify Sequences:

- **Tool:** Classify.seqs
- **Purpose:** Assigns a taxonomic classification (e.g., "Bacteria;Verrucomicrobia;...") to each sequence.
- **Key Parameters:**
 - fasta: The .fasta file from Pre.cluster.
 - reference: The trainset...fasta file.
 - taxonomy: The trainset...tax file.
 - count: The count table from Pre.cluster.

4. Cluster into OTUs:

- **Tool:** Cluster.split
- **Purpose:** Groups sequences into OTUs based on their taxonomic assignment and sequence similarity.
- **Key Parameters:**
 - Split by: Classification using fasta
 - fasta: The .fasta file from Pre.cluster.
 - taxonomy: The .taxonomy file from Classify.seqs.
 - count: The count table from Pre.cluster.
 - cutoff: 0.15 (a large distance for splitting).

5. Create Final OTU Table:

- **Tool:** Make.shared
- **Purpose:** Creates the final OTU abundance table (rows=samples, columns=OTUs, cells=counts).
- **Key Parameters:**
 - list: The .list file from Cluster.split.
 - name file or count table: The count table from Pre.cluster.
 - label: 0.03 (This defines the final OTU clustering level at 97% similarity).

6. Visualization

- **Tool:** Krona
- **Purpose:** Creates an interactive pie chart to visualize the taxonomic breakdown of the community.
- **Input:** The .tax.summary file (generated by Classify.seqs).

